El-Houcine Oussi

Software Engineer · Machine Learning

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SUMMARY

Software engineer with an engineering degree in information systems and two internships building Flutter, ASP.NET Core and Django applications. Now doing an M.Sc. at NUIST, where I build and evaluate solar power forecasting models in Python (LightGBM, PyTorch). Looking for software engineering or machine learning engineering roles.

EXPERIENCE

Graduate Researcher, Solar Power Forecasting, *NUIST* Sep 2024 – present · Nanjing

- Build data pipelines and forecasting models for photovoltaic (PV) power, from minutes to a day ahead, with pandas, NumPy, pvlib, LightGBM and PyTorch.
- Compared LightGBM, a GRU and a compact Transformer at 2–12 h lead times on PV systems in Australia, Hong Kong and the US. For the US system, the evaluation plan was fixed and its file hashes recorded before any forecast was made.
- Released the code and aggregate results under the MIT licence (GitHub, DOI 10.5281/zenodo.23590205). A verification script checks every file against a SHA-256 manifest; re-running the release reproduced all published tables exactly.
- Prepare Himawari / Meteosat satellite imagery and numerical weather forecasts (GRIB, ERA5) as model inputs for a study of 50 Australian solar farms.

Backend & Mobile Developer Intern, *Majal Berkane (Smart City)* Mar 2024 – Jul 2024 · Berkane, Morocco

- Built a Flutter mobile app using MVVM with Riverpod for state management.
- Wrote REST APIs in ASP.NET Core following Clean Architecture and the repository pattern.
- Tuned database queries and added server-side caching on the endpoints serving sensor data.

Web Development Intern, *Provincial Directorate of Equipment, Transport and Logistics* Mar 2023 – Jul 2023 · Oujda, Morocco

- Built an internal Django application for staff records, equipment scheduling and administrative logistics.
- Added role-based access control and report generation with pandas and NumPy.
- Deployed it on Windows Server with IIS and set up scheduled SQL Server backups.

PROJECTS

Connect Check · Flutter, Firebase Realtime Database, Riverpod, SQLite 2024 – 2025

- App where users join a shared room and their answers sync in real time through Firebase Realtime Database; includes localisation and an offline SQLite cache. Web version: lovecheckapp.web.app

MystoGuess · Flutter, Dart, SQLite 2025

- Guessing game with category challenges, solo mode and turn-based multiplayer, published on Google Play under the SystemX developer name. Word lists are stored locally in SQLite and JSON.

EcoBeach · Flutter, ASP.NET Core, C# 2024

- Mobile app and backend for booking beach services: sign-up and login, a prepaid wallet, reservations, and invoices with QR codes. Design report on oussi108.github.io.

Task API and expression compiler · C#, ASP.NET Core, EF Core, SQL Server 2023 – 2024

- Task-management REST API with JWT authentication, refresh tokens and EF Core migrations (taskwithauthdotnet); a math expression compiler with a tokenizer, recursive-descent parser and AST evaluator (Complier_EXP_math).

EDUCATION

M.Sc., Computer Science and Technology, *Nanjing University of Information Science and Technology* Sep 2024 – Jun 2027 (expected)

State Engineering Degree, Computer Science and Information Systems, *SUPMTI, Oujda, Morocco* Sep 2021 – Jun 2024

Specialized Technician Diploma, Software Development, *ISTA Lazaret, Oujda, Morocco* Sep 2019 – Jun 2021

SKILLS

Languages	Python, C#, Dart, Java, TypeScript, JavaScript, SQL, Bash
Backend & web	ASP.NET Core, Entity Framework Core, Django, Spring Boot, Angular, Ionic, React, REST APIs, JWT
Mobile	Flutter (Riverpod, MVVM), Firebase (Auth, Firestore, Realtime Database), Google Play publishing
ML & data	PyTorch, LightGBM, scikit-learn, pandas, NumPy, pvlib, Jupyter, satellite and NWP (GRIB) data
Data & ops	SQL Server, PostgreSQL, MySQL, SQLite, MongoDB, Docker, Git, GitHub Actions, Linux, IIS
Spoken	Arabic and Tamazight (native), English (fluent), French (professional), Mandarin Chinese (HSK 3)